$4\pi\varepsilon_0 r$

20. Answer: **D**

$$a = \frac{F_E}{m} = \frac{Ee}{m}$$

Applying $v^2 = u^2 + 2as$ in the vertical direction,

$$v_y^2 = 0 + 2\frac{Ee}{m}y$$

$$v_y = \sqrt{\frac{2Eey}{m}}$$

$$v_{\text{final}}^2 = v_x^2 + v_y^2 = v^2 + \frac{2Eey}{m}$$

$$v_{\text{final}} = \sqrt{v^2 + \frac{2Eey}{m}}$$

21. Answer: **B**